

The QFT is

$$|j\rangle \rightarrow \frac{1}{\sqrt{2^n}} \sum \exp\left(2\pi i j \frac{k}{2^n}\right) |k\rangle$$

After apply QFT, we have

$$\frac{1}{\sqrt{R}} \sum_q |qr\rangle \rightarrow \frac{1}{\sqrt{R2^n}} \sum_k \left[\sum_q \exp\left(2\pi i r k \frac{q}{2^n}\right) \right] |k\rangle$$

And we have

$$\sum_{q=0}^{R-1} \exp\left(2\pi i r k \frac{q}{2^n}\right) = \exp\left(\pi i (R-1) r \frac{k}{2^n}\right) \frac{\sin\left(\pi R r \frac{k}{2^n}\right)}{\sin\left(\pi r \frac{k}{2^n}\right)}.$$

Thus, the probability of measuring outcome k is

$$p(k | r) = \frac{1}{R2^n} \left| \sum_{q=0}^{R-1} \exp\left(2\pi i r k \frac{q}{2^n}\right) \right|^2 = \frac{1}{R2^n} \left(\frac{\sin\left(\pi R r \frac{k}{2^n}\right)}{\sin\left(\pi r \frac{k}{2^n}\right)} \right)^2.$$

Let $\varepsilon := \frac{r}{2^n}$, $\delta := \frac{1}{2^n}$, $x = \frac{\pi k}{2^n}$. In pratical, $r \ll 2^n$, thus ε is a small number. Thus, here we can take the continuous limit and approximate the probability density function as

$$f(x | r) = \frac{1}{\pi R} \left(\frac{\sin(Rrx)}{\sin(rx)} \right)^2.$$

we can verify that

$$\int_0^\pi f(x | r) dx = 1.$$

The Fisher information is

$$I(r) = E_r \left[\left(\frac{\partial}{\partial r} \ln f(x|r) \right)^2 \right]$$

$$\ln f(x|r) = -\ln(\pi R) + 2 \ln(\sin(Rrx)) - 2 \ln(\sin(rx))$$

Taking the derivative with respect to r (score function):

$$\frac{\partial}{\partial r} \ln f(x|r) = 2Rx \cot(Rrx) - 2x \cot(rx)$$

$$\frac{\partial^2}{\partial r^2} \ln f(x|r) = -2R^2 x^2 (1 + \cot^2(Rrx)) + 2x^2 (1 + \cot^2(rx))$$

Taking the expectation:

$$\begin{aligned} I(r) &= \int_0^\pi f(x|r) [2R^2 x^2 (1 + \cot^2(Rrx)) - 2x^2 (1 + \cot^2(rx))] dx \\ &= \frac{2}{\pi R} \int_0^\pi x^2 \left[\frac{R^2}{\sin^2(rx)} - \frac{\sin^2(Rrx)}{\sin^4(rx)} \right] dx \end{aligned}$$

The Fejer kernel gives

$$\begin{aligned}\left(\frac{\sin(\frac{nx}{2})}{\sin(\frac{x}{2})}\right)^2 &= \sum_{k=-(n-1)}^{n-1} (n - |k|) \exp(ikx) \\ &= n + 2 \sum_{k=1}^{n-1} (n - k) \cos(kx)\end{aligned}$$

Let $n = R$, $x = 2rx$, we have

$$\left(\frac{\sin(Rrx)}{\sin(rx)}\right)^2 = R + 2 \sum_{k=1}^{R-1} (R - k) \cos(2krx)$$

Plugging into the Fisher information, we have

$$I(r) = \frac{2}{\pi R} \int_0^\pi \frac{x^2}{\sin(rx)^2} \left[R^2 - \left(R + 2 \sum_{k=1}^{R-1} (R - k) \cos(2krx) \right) \right] dx$$

等差数列求和给出

$$R^2 - R = \sum_{k=1}^{R-1} 2(R - k)$$

then

$$\begin{aligned}I(r) &= \frac{4}{\pi R} \int_0^\pi \frac{x^2}{\sin(rx)^2} \sum_{k=1}^{R-1} 2(R - k) \sin^2(krx) dx \\ &= \frac{8}{\pi R} \sum_{k=1}^{R-1} (R - k) \int_0^\pi x^2 \left(\frac{\sin(krx)}{\sin(rx)} \right)^2 dx\end{aligned}$$

Plugging the Fejer kernel again, we have

$$I(r) = \frac{8}{\pi R} \sum_{k=1}^{R-1} (R - k) \int_0^\pi x^2 \left[k + 2 \sum_{j=1}^{k-1} (k - j) \cos(2jrx) \right] dx$$

We know that

$$\int_0^\pi x^2 dx = \frac{\pi^3}{3}$$

$$\int_0^\pi x^2 \cos(2jrx) dx = \frac{1}{(2jr)^3} [4\pi jr \cos(2\pi jr) + (4\pi^2 j^2 r^2 - 2) \sin(2\pi jr)]$$

Thus,

$$I(r) = \frac{8}{\pi R} \sum_{k=1}^{R-1} (R - k) \left[\frac{k\pi^3}{3} + \sum_{j=1}^{k-1} \frac{k-j}{2j^3 r^3} (2\pi jr \cos(2\pi jr) + (2\pi^2 j^2 r^2 - 1) \sin(2\pi jr)) \right]$$

Notice that the r is a big number, $\frac{1}{r} \ll 1$, so we throw out the high order small terms, we have

$$I(r) \approx \frac{8\pi^2}{3R} \sum_{k=1}^{R-1} k(R - k) = \frac{4\pi^2}{9} (R^2 - 1)$$